

Role of Bicarbonate Ions in Aqueous Solution as a Carbon Source for Photocatalytic Conversion of CO₂ into CO

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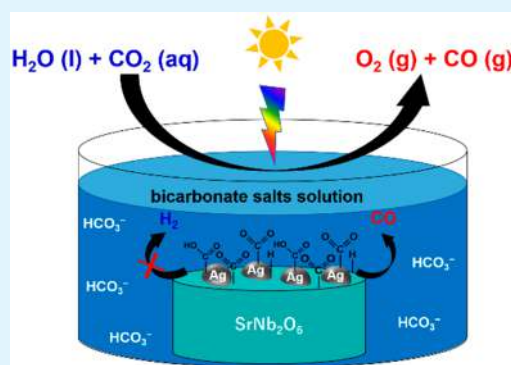
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S Supporting Information

ABSTRACT: Ag/SrNb₂O₆ nanorods exhibited a high CO formation rate and high selectivity toward CO evolution in aqueous solutions containing bicarbonate ions even without CO₂ bubbling. Notably, the formation rate of CO reached as high as 287 μmol h⁻¹ with selectivity toward CO evolution of higher than 94.1% when NH₄HCO₃ was used as an additive. Both the formation rate of CO and selectivity toward CO evolution increased with the concentration of HCO₃⁻. According to the experimental results and analytical chemistry calculations, it was concluded that CO₂(aq) obtained by the dissociation of HCO₃⁻ is the actual reactant in the photocatalytic conversion of CO₂. In contrast, the HCO₃⁻ species in the aqueous solution is beneficial for improving the photocatalytic activity and selectivity toward CO evolution by increasing the adsorption of carbon-related species on the photocatalyst surface and/or suppressing the backward reaction for the photocatalytic conversion of CO₂.

KEYWORDS: photocatalysis, bicarbonate, carbon dioxide, NH₄HCO₃, SrNb₂O₆



INTRODUCTION

The “Climate Change 2014: Synthesis Report” by the Intergovernmental Panel on Climate Change predicted that anthropogenic greenhouse gas (GHG) emissions since the preindustrial era have driven large increases in atmospheric concentrations of carbon dioxide (CO₂). Without additional efforts to reduce GHG emissions, the environmental CO₂ concentration will exceed 450 ppm of CO₂-eq by 2030 and reach between ~750 and >1300 ppm of CO₂-eq by 2100, which can have a disastrous impact on the global climate, environment, and economy.¹ Inspired by plant photosynthesis, one of the best strategies to mitigate CO₂ emissions is the photocatalytic conversion of CO₂ into other feedstocks such as CO, HCOOH, HCHO, CH₄, and CH₃CH₂OH using heterogeneous catalysts and H₂O as an electron donor.^{2–6} Especially, the conversion of CO₂ into CO can be further used for syngas preparation based on the Fischer–Tropsch process.^{7–9} Nevertheless, it is difficult to activate CO₂ selectively and suppress the H₂ evolution from protons (H⁺) in aqueous solutions, because the redox potential of CO₂/CO (−0.521 V vs SHE, pH = 7) is more negative than that of H⁺/H₂ (−0.414 V vs SHE, pH = 7).¹⁰ To our knowledge, Kudo et al. reported for the first time that the formation of CO from CO₂ exceeds that of H₂ from H⁺, and a stoichiometric amount of O₂ evolves as an oxidation product for the photocatalytic conversion of CO₂ in an aqueous solution over Ag/

BaLa₄Ti₄O₁₅.¹¹ Subsequently, various Ag-loaded photocatalysts have been reported for the highly selective photocatalytic conversion of CO₂ into CO with H₂O as an electron donor.^{12–15} Our group also developed many photocatalysts that exhibit a high selectivity toward CO evolution for the photocatalytic conversion of CO by H₂O, such as Ag-modified ZnGa₂O₄/Ga₂O₃,^{16,17} ZnGa₂O₄,¹⁸ Sr₂KTa₅O₁₅,¹⁹ Mg–Al LDH/Ga₂O₃,²⁰ K₂RETa₅O₁₅,²¹ and Ag–Cr-modified Ga₂O₃.²²

According to previous isotopic experiments and in situ FT-IR spectroscopy, CO₂ dissolved in an aqueous solution (CO₂(aq)) will react with the hydroxyl group anchored on the surface to form bidentate bicarbonate, which acts as the intermediate species for the photocatalytic conversion of CO₂, wherein H₂O acts as the electron donor.²³ We found that CO₂(aq) functions as the reactant for the photocatalytic conversion of CO₂ to CO, and its concentration ([CO₂(aq)]) significantly influences the formation rate of CO and selectivity toward CO evolution. In our previous work,^{16–19,21,23–25} the main reduction product was found to change from CO to H₂ derived from water splitting when [CO₂(aq)] was too low. In fact, the solubility of CO₂ in pure H₂O is only ~0.033 mol L⁻¹ (at 298 K under 1 atm).^{26,27} It is extremely difficult to achieve

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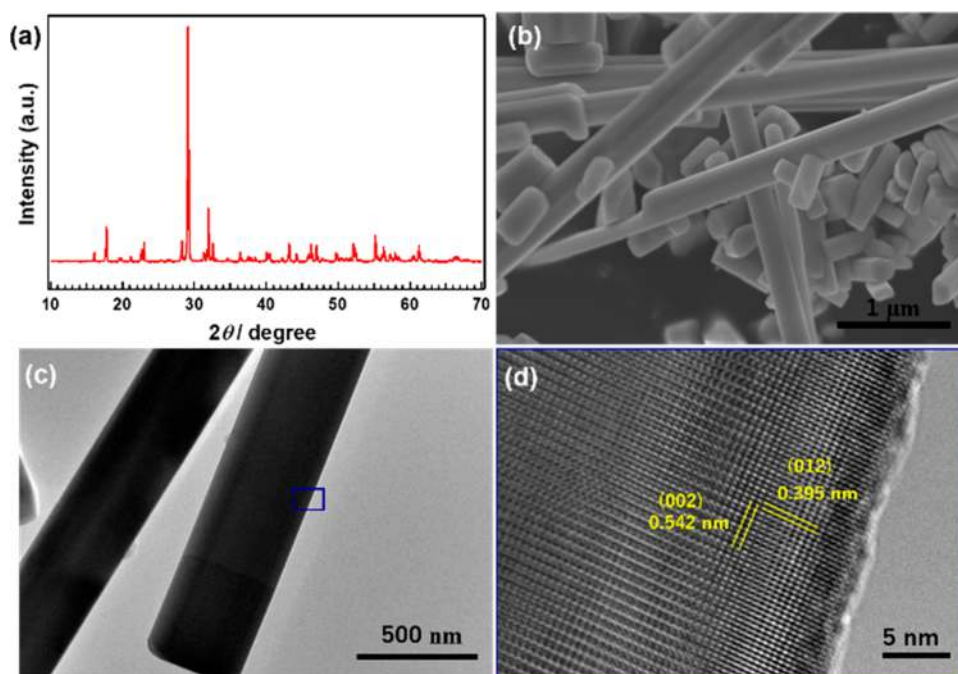


Figure 1. (a) XRD pattern, (b) SEM image, (c) TEM image, and (d) high-resolution TEM (HRTEM) image of SrNb_2O_6 ; (d) shows an enlarged HRTEM image of the area enclosed in the blue rectangle in (c).

efficient photocatalytic conversion of CO_2 using H_2O as an electron donor at low concentrations of CO_2 . Moreover, the capture, storage, and transportation of CO_2 are very expensive. Thus, it is crucial to develop a photocatalyst that can facilitate highly selective and highly active photocatalytic conversion of CO_2 to CO at low CO_2 concentrations. In this article, we propose a strategy for using bicarbonate as the carbon source for the photocatalytic conversion of CO_2 by H_2O over $\text{Ag}/\text{SrNb}_2\text{O}_6$. The formation rate of CO reached as high as $287 \mu\text{mol h}^{-1}$ with selectivity toward CO evolution of higher than 94.1% when NH_4HCO_3 was used as a carbon source. Furthermore, the intermediate species for the photocatalytic conversion of CO_2 with bicarbonate as the carbon source were evaluated through experiments and analytical chemistry calculations.

MATERIALS AND EXPERIMENTAL METHODS

Preparation of the Photocatalyst. SrNb_2O_6 was prepared by a flux method. Briefly, 2.0 g of Nb_2O_5 powder (99.9%, Wako) and 6.0 g of $\text{SrCl}_2 \cdot 6\text{H}_2\text{O}$ (99.9%, Wako) were ground in an alumina mortar for 5 min. The mixture was calcined in air using an alumina crucible at 1173 K for 2 h. After calcination, the obtained powder was thoroughly washed three times with hot water (353 K) to remove the residual salt and dried at 353 K in an oven.²⁸ The impregnation method was used to load 1.0 wt % of Ag on the surface of the SrNb_2O_6 photocatalyst. Specifically, 1.5 g of the prepared SrNb_2O_6 was homogeneously dispersed in an aqueous AgNO_3 solution (20 mL), water was removed by evaporation at 358 K, and then the product was calcined at 723 K for 2 h in air.

Photocatalyst Characterization. The crystal phase of SrNb_2O_6 was characterized by powder X-ray diffractometry (XRD; Rigaku Multiflex) with $\text{Cu K}\alpha$ radiation ($\lambda = 0.154 \text{ nm}$). The morphology of the sample was observed by field-emission scanning electron microscopy (FE-SEM, SU8220, Hitachi High-Technologies) and transmission electron microscopy (TEM, JEM-2100F, JEOL). The adsorbed carbon species were characterized by Fourier transform infrared (FTIR) spectroscopy performed using an FTIR spectrometer (Model: FT/IR-4700, JASCO International Co., Ltd., Japan)

equipped with a mercury–cadmium–tellurium detector and cooled by liquid N_2 in transmission mode at 303 K. Each sample ($\sim 30 \text{ mg}$) was pressed into a wafer (diameter: 10 mm) and introduced into the instrument in a cylindrical glass cell with calcium fluoride (CaF_2) windows. The wafer was evacuated at 673 K for 30 min before measurements, followed by treatment with $\sim 40 \text{ Torr}$ of O_2 for 30 min, and the wafer was subsequently evacuated for 30 min and cooled to 303 K. The data for each FTIR spectrum were obtained from 128 scans with a resolution of 4 cm^{-1} .

Photocatalytic Activity Test for the Conversion of CO_2 with H_2O . The photocatalytic conversion of CO_2 was carried out using a flow system with an inner-irradiation-type reaction vessel at ambient pressure; a schematic of the reactor is illustrated in Figure S1. In ultrapure water (1.0 L) containing a certain concentration of additives, 0.5 g of the synthesized photocatalyst was dispersed. CO_2 and/or Ar gas were bubbled into the solution continuously at a flow rate of 30 mL min^{-1} . The suspension was illuminated using a 400-W high-pressure mercury lamp with a quartz filter connected to a cooling water system. The amounts of the evolved H_2 , O_2 , and N_2 were detected using a thermal conductivity detector–gas chromatography (GC) system (Shimadzu Corp; MS-5A column, Ar carrier), and the amount of evolved CO was analyzed by a flame ionization detector–GC with a methanizer (ShinCarbon ST column, N_2 carrier). The selectivity toward CO evolution compared to the H_2 evolution and the balance between the consumed electrons (e^-) and holes (h^+) can be expressed by eqs 1 and 2, respectively.²

$$\begin{aligned} \text{Selectivity toward CO evolution (\%)} \\ = 100 \times 2R_{\text{CO}} / (2R_{\text{CO}} + 2R_{\text{H}_2}) \end{aligned} \quad (1)$$

$$\text{Consumed } e^-/h^+ = (2R_{\text{CO}} + 2R_{\text{H}_2}) / 4R_{\text{O}_2} \quad (2)$$

Here, R_{CO} and R_{H_2} represent the formation rates of CO and H_2 , respectively.

In the backward reaction, the processes were almost the same as those of the photocatalytic reduction of CO_2 , except that CO, O_2 , and diluent gas Ar were bubbled into ultrapure water and the NaHCO_3 solution, respectively, at a total flow rate of 30 mL min^{-1} , and the amount of evolved CO_2 was analyzed by FID-GC. In order to

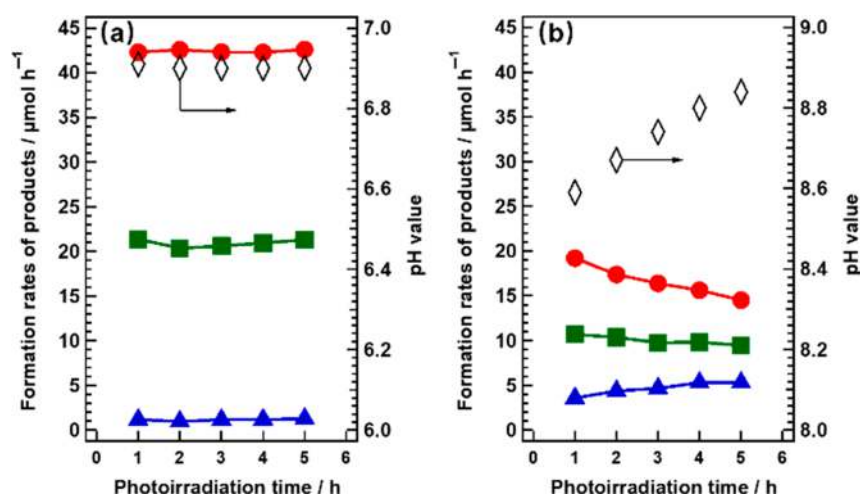


Figure 2. Formation rates of H_2 (blue triangle), O_2 (green square), CO (red circle), and pH of the solution (open diamond) in the photocatalytic conversion of CO_2 in a NaHCO_3 aqueous solution over $\text{Ag}/\text{SrNb}_2\text{O}_6$ with the continuously bubbling of (a) CO_2 and (b) Ar gas. Photocatalyst: 1.0 wt % $\text{Ag}/\text{SrNb}_2\text{O}_6$ (0.5 g); reaction volume: H_2O (1.0 L); gas flow rate: 30 mL min^{-1} ; light source: 400-W high-pressure Hg lamp.

eliminate the influence of the Ag cocatalyst, bare SrNb_2O_6 was used as the photocatalyst in the backward reaction.

RESULTS AND DISCUSSION

The crystal structures of SrNb_2O_6 were confirmed by powder XRD analysis, as illustrated in Figure 1a. All diffraction peaks in the XRD pattern can be precisely indexed to pure monoclinic SrNb_2O_6 with the space group $\text{P}12_1/\text{c}1$ (JCPDS 01-072-2088).²⁹ SEM and TEM images of SrNb_2O_6 are shown in Figure 1b,c, respectively, which indicate that the produced SrNb_2O_6 mainly consists of a 1D nanorod structure. The corresponding high-resolution TEM image in Figure 1d shows fringe spacings of 0.395 and 0.542 nm, which correspond to the (012) and (002) lattice planes of monoclinic SrNb_2O_6 , respectively. These results indicate that the SrNb_2O_6 nanorods are of good crystallinity with a growth direction parallel to the (012) plane of the lattice.

Figure 2a shows the formation rates of H_2 , O_2 , and CO during the photocatalytic conversion of CO_2 for 5 h with the bubbling of CO_2 into the NaHCO_3 solution. The formation rate of CO is $\sim 42.3 \mu\text{mol h}^{-1}$ and is very stable during the photoirradiation for 5 h. A very small amount of H_2 is formed, and the selectivity toward CO evolution is higher than 97.3%. The pH of the reaction solution remained stable at 6.86 during the photocatalytic conversion of CO_2 for 5 h during continuous bubbling of CO_2 . However, upon flowing Ar gas instead of CO_2 gas, as shown in Figure 2b, the formation rate of CO decreased by approximately half to $19.2 \mu\text{mol h}^{-1}$, and the formation rate of H_2 increased slightly after 1 h of photoirradiation. With the increase in the photoirradiation time, the pH of the solution increased from 8.58 to 8.89, and the formation rate of CO decreased while that of H_2 increased after photoirradiation for 5 h under the flow of Ar.

Various control experiments were conducted to confirm the CO_2 source under the flow of Ar. No products were detected in the dark (Figure S2a) and without a photocatalyst (Figure S2b). Table 1 lists the formation rates of products in the photocatalytic conversion of CO_2 over $\text{Ag}/\text{SrNb}_2\text{O}_6$ with different additives under the bubbling of Ar. While H_2 and O_2 were observed as the main products, the formation rate of CO was negligible without any additives (Entry 1). This result indicates that additives are indispensable for the photocatalytic

Table 1. Photocatalytic Conversion of CO_2 over $\text{Ag}/\text{SrNb}_2\text{O}_6$ with Different Additives under the Bubbling of Ar Gas^a

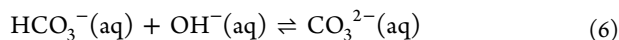
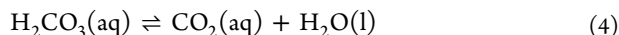
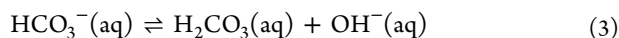
entry	additives	pH	formation rates of gases ($\mu\text{mol h}^{-1}$)			selec. toward CO (%)
			H_2	O_2	CO	
1	none	6.90	2.9	1.2	0.1	3.3
2	NaHCO_3^b	8.59	3.6	10.7	19.2	87.8
3	KHCO_3^b	8.56	2.2	11.2	20.6	90.3
4	$\text{NH}_4\text{HCO}_3^b$	8.28	17.9	94.2 (N_2)	287	94.1
5	Na_2CO_3^c	11.21	14.9	7.7	0.1	0.7
6	NaOH^b	12.89	17.7	8.2	0.3	1.5
7	$\text{Na}_2\text{HPO}_4/\text{NaH}_2\text{PO}_4$	8.30	18.3	9.1	0.1	0.5
8	$\text{Na}_2\text{CO}_3 + \text{CO}_2^d$	8.51	2.0	13.1	26.3	93.0

^aPhotocatalyst: 1.0 wt % $\text{Ag}/\text{SrNb}_2\text{O}_6$ (0.5 g); reaction volume: H_2O (1.0 L); Ar flow rate: 30 mL min^{-1} ; light source: 400-W high-pressure Hg lamp. ^bAdditive concentration 0.1 M. ^cAdditive concentration 0.05 M. ^d CO_2 was bubbled into 0.05 M Na_2CO_3 aqueous solution until the pH of the reaction solution reached 8.51, and then the CO_2 gas was switched to Ar gas.

conversion of CO_2 over $\text{Ag}/\text{SrNb}_2\text{O}_6$ in an aqueous solution. A high selectivity toward CO evolution is obtained in the case of NaHCO_3 , KHCO_3 , and NH_4HCO_3 as additives (Entries 2–4). Notably, a very high formation rate of CO and selectivity toward CO evolution are achieved with NH_4HCO_3 as an additive under the bubbling of Ar gas instead of CO_2 gas ($287 \mu\text{mol h}^{-1}$ and 94.2%, respectively). In NaHCO_3 aqueous solution, H_2O was used as an electron donor and oxidized to O_2 , while in NH_4HCO_3 solution, NH_4^+ and NH_3 will be generated, which can be readily oxidized to N_2 during the photocatalytic conversion of CO_2 . The standard Gibbs free energy change ΔG^0 for the decomposition of NH_3 to N_2 (18 kJ mol^{-1})³⁰ is considerably smaller than that required for the decomposition of H_2O to O_2 (237 kJ mol^{-1}). This implies that the photocatalytic oxidation of NH_4^+ and NH_3 is more favorable than the oxidation of H_2O to O_2 . Consequently, the activity toward CO evolution is higher in NH_4HCO_3 solution than that in NaHCO_3 solution. Moreover, a stoichiometric

amount of N_2 is obtained as the oxidation product, indicating that ammonia and/or ammonium ions function as electron donors in the photocatalytic conversion of CO_2 to CO; the details of calculations carried out according to our previous work are shown in the SI.³¹ A very small amount of CO evolved upon using other additives, such as Na_2CO_3 , NaOH, and a mixed solution of Na_2HPO_4/NaH_2PO_4 , which show similar pH values as in the $NaHCO_3$ solution (Entries 5–7). If CO_2 is bubbled into 0.05 M Na_2CO_3 until the pH of the reaction solution at 8.51 is close to that of the $NaHCO_3$ solution at 8.59 and then changed to Ar gas, comparable amounts of CO, H_2 , and O_2 will be obtained as compared to those in the case of $NaHCO_3$ in the photocatalytic conversion of CO_2 under the bubbling of Ar. According to the carbon equilibrium, after bubbling CO_2 into 0.05 M Na_2CO_3 until the pH reaches 8.51, the final solution will be similar to 0.1 M $NaHCO_3$; therefore, the major species of dissolved CO_2 is HCO_3^- .³² This result indicates that HCO_3^- acts as the carbon source in the photocatalytic conversion of CO_2 into CO in the aqueous solution containing bicarbonate ions under the bubbling of Ar.

HCO_3^- is converted into $H_2CO_3(aq)$ because there are no CO_2 -related species in the aqueous solution. $H_2CO_3(aq)$ will further produce $CO_2(aq)$ easily because of the low hydration equilibrium constant of carbonic acid in pure water.²³ Based on the carbon equilibrium in the solution, there are primarily four equilibria in the solution, as follows:



Based on eqs 3 and 6, the first and second dissociation constants of carbonic acid can be denoted by eqs 7 and 8, respectively:

$$K_1 = \frac{[OH^-][H_2CO_3]}{[HCO_3^-]} = \frac{K_w[H_2CO_3]}{[H^+][HCO_3^-]} \quad (7)$$

$$K_2 = \frac{[CO_3^{2-}]}{[HCO_3^-][OH^-]} = \frac{[H^+][CO_3^{2-}]}{K_w[HCO_3^-]} \quad (8)$$

where $K_w = 1.0 \times 10^{-14}$ is the self-ionization constant of water:

$$K_w = [H^+][OH^-] \quad (9)$$

As mentioned in our previous paper, because the CO_2 hydration equilibrium constant is very small, the $[H_2CO_3]$ mentioned in eqs 3 and 4 is almost equal to $[CO_2(aq)]$,²³ and $K_1' = K_w/K_1$, $K_2' = K_2 \times K_w$. Thus, eqs 7 and 8 can be changed to

$$K_1' = \frac{[H^+][HCO_3^-]}{[CO_2(aq)]} \quad (10)$$

$$K_2' = \frac{[H^+][CO_3^{2-}]}{[HCO_3^-]} \quad (11)$$

In the $NaHCO_3$ aqueous solution, there are primarily three kinds of CO_2 -related species: $CO_2(aq)$, HCO_3^- , and CO_3^{2-} . Herein, we define D as the total concentration of all CO_2 -related species.

$$D = [CO_2(aq)] + [HCO_3^-] + [CO_3^{2-}] \quad (12)$$

Substituting eqs 10 and 11 into eq 12 results in

$$[HCO_3^-] = \frac{K_1'[H^+]}{[H^+]^2 + K_1'[H^+] + K_1'K_2'} \cdot D \quad (13)$$

$$[CO_3^{2-}] = \frac{K_1'K_2'}{[H^+]^2 + K_1'[H^+] + K_1'K_2'} \cdot D \quad (14)$$

$$[CO_2(aq)] = \frac{[H^+]^2}{[H^+]^2 + K_1'[H^+] + K_1'K_2'} \cdot D \quad (15)$$

In contrast, according to the ionization balance of ions in the $NaHCO_3$ solution, which exhibits electrical neutrality,

$$[Na^+] + [H^+] = [HCO_3^-] + 2[CO_3^{2-}] + [OH^-] \quad (16)$$

When eqs 13, 14, and 15 are introduced into eq 16, the value of D can be obtained as

$$D = \left([Na^+] + [H^+] - \frac{K_w}{[H^+]} \right) \left(\frac{[H^+]^2 + K_1'[H^+] + K_1'K_2'}{K_1'[H^+] + 2K_1'K_2'} \right) \quad (17)$$

Plummer and Busenberg³³ reported that the dissociation constants of carbonic acid, K_1' and K_2' , can be calculated from the temperature of the solution; the empirical expression is as follows:

$$\begin{aligned} \log K_1' &= -356.3094 - 0.06091964T + \frac{21834.37}{T} \\ &+ 126.8339 \log T - \frac{1684915}{T^2} \end{aligned} \quad (18)$$

$$\begin{aligned} \log K_2' &= -107.8871 - 0.03252849T + \frac{5151.79}{T} \\ &+ 38.92561 \log T - \frac{563713.9}{T^2} \end{aligned} \quad (19)$$

where T is the temperature of the solution.

Therefore, the concentrations of all the CO_2 -related species ($CO_2(aq)$, HCO_3^- , and CO_3^{2-}) in a carbonic acid buffer aqueous solution can be estimated by the temperature and pH of the solution. In this work, the measured pH values ranged from 4.0 to 12.0 during the photocatalytic conversion of CO_2 . Figure S3 presents the calculated concentrations of $CO_2(aq)$, HCO_3^- , and CO_3^{2-} in a 0.10 M aqueous solution of $NaHCO_3$ at 303 K under 101.325 kPa of CO_2 . During the photocatalytic conversion of CO_2 in a $NaHCO_3$ aqueous solution with the bubbling of Ar gas, the pH values of the solutions are approximately 8.0–9.0. In this pH range, $[HCO_3^-]$ is almost stable at 0.10 M; $[CO_2(aq)]$ is quite low, ranging from 2.11×10^{-3} to 1.93×10^{-4} M; and $[CO_3^{2-}]$ ranges from 5.09×10^{-4} to 4.66×10^{-3} M. In our previous work, we proved that $CO_2(aq)$ is the direct reactant for the photocatalytic conversion of CO_2 with H_2O as an electron donor over $Ag/ZnGa_2O_4/Ga_2O_3$ with the continuous bubbling of CO_2 .²³ If Ar was bubbled instead of CO_2 , H_2 would become the main product and the CO evolved would be negligible because of the low $[CO_2(aq)]$. On the contrary, the selectivity toward CO evolution was still very high in this work even though $[CO_2(aq)]$ was very low. The adsorption of CO_2 on the $ZnGa_2O_4/Ga_2O_3$ and $SrNb_2O_6$ surfaces was measured using in situ FTIR spectroscopy (Figure S4). The bands assigned to

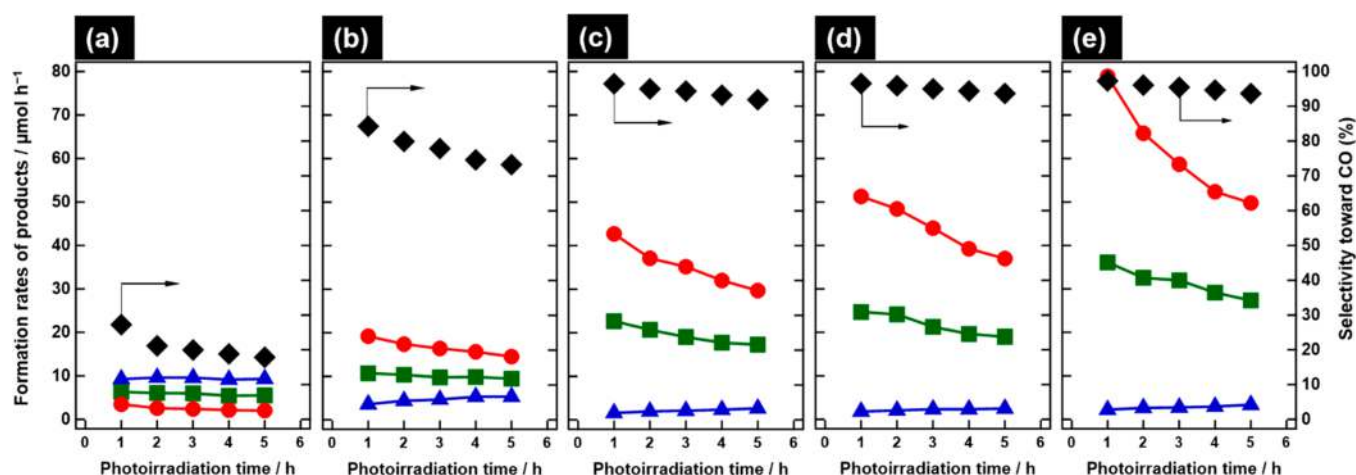


Figure 3. Formation rates of H₂ (blue triangles), O₂ (green squares), and CO (red circles) and selectivity toward CO evolution (black diamonds) at different concentrations of NaHCO₃: (a) 0.01, (b) 0.10, (c) 0.30, (d) 0.50, and (e) 1.00 M for the photocatalytic conversion of CO₂ after photoirradiation for 5 h. Photocatalyst: 1.0 wt % Ag/SrNb₂O₆ (0.5 g); reaction volume: H₂O (1.0 L); Ar flow rate; 30 mL min⁻¹; light source: 400-W high-pressure Hg lamp.

CO₂ adsorbed on ZnGa₂O₄/Ga₂O₃ are much higher than those of CO₂ adsorbed on SrNb₂O₆. Therefore, we considered that the good adsorption for CO₂ on the surface of the photocatalyst is not the key factor leading to the high CO evolution at low CO₂(aq) concentration. As we found in our previous work, the Ag cocatalyst nanoparticles were selectively deposited on specific surface sites owing to the anisotropy of SrNb₂O₆ nanorods under photoirradiation, as shown in Figure S5.²⁸ The separation of the reduction and oxidation sites is considered to be crucial for the high photocatalytic selectivity toward CO evolution at low CO₂ (aq) concentrations over Ag/SrNb₂O₆.

In order to confirm which species is the actual reactant during the photocatalytic conversion of CO₂ under the flow of Ar in the aqueous solution of bicarbonate salt, we investigated the effect of the NaHCO₃ concentration ([NaHCO₃]) and the dependences of the formation rate of CO on [CO₂(aq)], [HCO₃⁻], and [CO₃²⁻].

Figure 3 shows the formation rates of H₂, O₂, and CO at different [NaHCO₃] in the photocatalytic conversion of CO₂ after photoirradiation for 5 h. When [NaHCO₃] is 0.01 M, the formation rate of CO is lower than that of H₂. The formation rate of CO and selectivity toward CO evolution steadily increases with [NaHCO₃] from 0.01 to 1.00 M. The formation rate of CO reaches 78.9 $\mu\text{mol h}^{-1}$ with selectivity toward CO evolution of higher than 97.3% after photoirradiation for 1 h in 1.00 M NaHCO₃ solution. It should be noted that the formation rate of H₂ decreases with [NaHCO₃], as shown in Figure S6. This result indicates that increasing the concentration of HCO₃⁻ is beneficial for improving the formation rate of CO and suppressing the formation of H₂. Notably, the formation rate of CO decreased dramatically with the reaction time at a high concentration of NaHCO₃; even when we carried out the cycle test for three times, the formation rate of CO also showed the same trend of decreasing with an increase in the photoirradiation time (Figure S7). Because we used a flow system for the photocatalytic conversion of CO₂ in our system, Ar gas was flowed into the reactor continuously, and the flowing Ar gas will carry away a part of the CO₂ gas derived from the dissociation of HCO₃⁻, which leads to an increase in the pH of the reaction solutions upon increasing the

photoirradiation time (Figure S8). According to eqs 13–19, the increase in the pH will result in a decrease in the [CO₂(aq)], especially when the concentration of NaHCO₃ is high.

Based on the pH values and reaction temperatures of different [NaHCO₃] solutions, we calculated the corresponding concentrations of carbon species (CO₂(aq), HCO₃⁻, and CO₃²⁻) derived from eqs 13, 14, and 15, according to eqs 17, 18, and 19. Figure 4 displays the dependences of the formation rate of CO on [CO₂(aq)], [HCO₃⁻], and [CO₃²⁻]. In a fixed concentration of NaHCO₃ solution, the formation rate of CO increases with [CO₂(aq)] and [HCO₃⁻], while it decreases

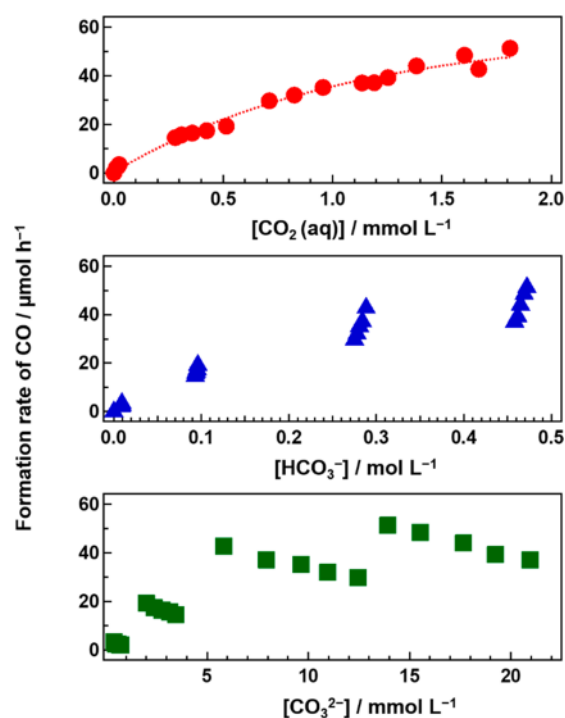


Figure 4. Dependences of the formation rate of CO on [CO₂(aq)] (red circles), [HCO₃⁻] (blue triangles), and [CO₃²⁻] (green squares) at different [NaHCO₃] ([NaHCO₃] = 0.01, 0.10, 0.30, and 0.50 M).

with $[\text{CO}_3^{2-}]$. As $[\text{NaHCO}_3]$ increases from 0.01 to 0.50 M, $[\text{CO}_2(\text{aq})]$, $[\text{HCO}_3^-]$, and $[\text{CO}_3^{2-}]$ increase steadily. However, in the whole $[\text{NaHCO}_3]$ range, the formation rate of CO only shows a good correspondence with $[\text{CO}_2(\text{aq})]$. This result clearly indicates that $\text{CO}_2(\text{aq})$ obtained by the dissociation of HCO_3^- is the actual reactant in the photocatalytic conversion of CO_2 , although $[\text{CO}_2(\text{aq})]$ is very low at solution pH values ranging from 8.0 to 9.0. Consequently, the yield of CO is defined as

$$Y = \frac{R_{\text{CO}}}{[\text{CO}_2(\text{aq})]} \quad (20)$$

Figure 2a shows that the formation rate of CO is $\sim 42.3 \mu\text{mol h}^{-1}$ with the pH of the reaction solution being stable at 6.86 when CO_2 gas is continuously bubbled into 0.10 M NaHCO_3 solution. The yield of CO is only $\sim 0.15\%$ in this case. Figure 5 shows the yield of CO in the photocatalytic

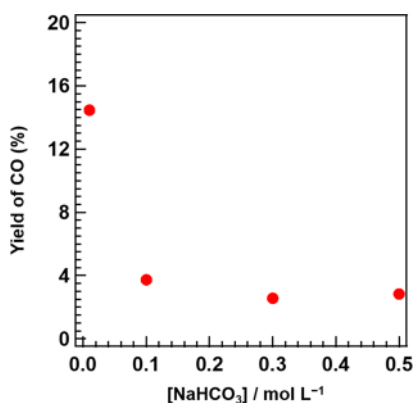


Figure 5. Yield of CO in the photocatalytic conversion of CO_2 with Ar gas bubbling at different $[\text{NaHCO}_3]$.

conversion of CO_2 with Ar gas bubbling at different $[\text{NaHCO}_3]$. The yield of CO is $\sim 3.0\%$ when $[\text{NaHCO}_3]$ ranges from 0.10 to 0.50 M and reaches $\sim 14.6\%$ when $[\text{NaHCO}_3]$ is 0.01 M, although H_2 is the main product in this case. The yield of CO in the photocatalytic conversion of CO_2 for 5 h in different $[\text{NaHCO}_3]$ solutions is shown in Figure S9, which indicates that the yield of CO increased slightly with an increase in the photoirradiation time, owing to the decrease in the $[\text{CO}_2(\text{aq})]$. This work suggests that using bicarbonate as a carbon source can significantly increase the conversion rate of CO_2 , and Ag/SrNb₂O₆ is a promising photocatalyst for the highly effective conversion of CO_2 to CO at low CO_2 concentrations.

We further investigated the dependence of the formation rate of CO on $[\text{CO}_2(\text{aq})]$ by varying the partial pressure of CO_2 in the gas phase. A change in the partial pressure of CO_2 will result in a change in the pH of the solution. According to eqs 15, 17, 18, and 19, $[\text{CO}_2(\text{aq})]$ can be calculated with different partial pressures of CO_2 . Figure 6 illustrates the dependence of the formation rate of CO on $[\text{CO}_2(\text{aq})]$ at different partial pressures of CO_2 . With the increase in the partial pressure of CO_2 , the pH of the solution gradually decreases, and the formation rate of CO increases exponentially with $[\text{CO}_2(\text{aq})]$ in the reaction solution, which is consistent with the result in Figure 4. This result further confirms that $\text{CO}_2(\text{aq})$ is the direct reactant in the photocatalytic conversion of CO_2 in the aqueous solution containing

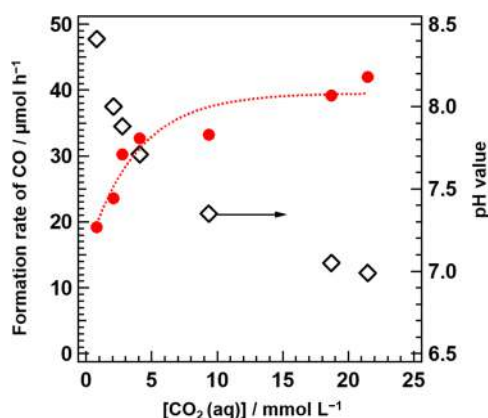


Figure 6. Dependence of formation rate of CO (red circle) and pH values (open diamond) on $[\text{CO}_2(\text{aq})]$ at different partial pressures of CO_2 . The partial pressure of CO_2 was adjusted by varying the flow rate of CO_2 and Ar in the gas phase of the fluid (flow rate ratios of CO_2 :Ar = 0:30, 2:28, 3:27, 5:25, 10:20, 20:10, and 30:0); the total flow rate of CO_2 and Ar gas was 30 mL min, $[\text{NaHCO}_3] = 0.1 \text{ M}$, and $T = 304.5 \text{ K}$.

bicarbonate ions. Note that the formation rate of CO did not increase remarkably when the concentration of $\text{CO}_2(\text{aq})$ was higher than 10 mmol h^{-1} , indicating that there is an upper limit of the amount of $\text{CO}_2(\text{aq})$ that can be adsorbed on the surface of the SrNb₂O₆ nanorod. This is because the specific surface area of SrNb₂O₆ is only $\sim 1.78 \text{ m}^2 \text{ g}^{-1}$,²⁸ and moreover, as shown in the SEM images in Figure S5, the Ag cocatalyst that acts as the active site for the reduction of CO_2 is selectively deposited on the edge of the SrNb₂O₆ nanorod under photoirradiation. This indicates that the amount of CO_2 adsorbed at the active sites is very limited. Therefore, when $[\text{CO}_2(\text{aq})]$ in the solution is increased to a certain concentration, it does not cause a significant increase in the CO formation.

Figure 7a,b shows the formation rates of H_2 , O_2 , and CO in the photocatalytic conversion of CO_2 in an aqueous solution of NaHCO_3 and in a pure H_2O solution, respectively. Because $\text{CO}_2(\text{aq})$ is considered to be the reactant in the photocatalytic conversion of CO_2 into CO, the partial pressure of CO_2 was adjusted to keep $[\text{CO}_2(\text{aq})]$ similar in two solutions. Almost no CO is evolved in the H_2O solution, while the formation rate of CO is as high as $30.2 \mu\text{mol h}^{-1}$ with selectivity higher than 89% in the NaHCO_3 solution. This result indicates that HCO_3^- has a significant influence on the photocatalytic activity and selectivity for the conversion of CO_2 , although it is not the direct reactant in the photocatalytic conversion of CO_2 .

Regarding the role of HCO_3^- , we speculate two possibilities: (1) As shown in Scheme 1, the primary adsorbed species on the photocatalyst is a carbonate or bicarbonate species derived from HCO_3^- , for pH ranging from 8.0 to 9.0 in an aqueous solution of NaHCO_3 with the flow of Ar gas. However, hydrogen species are mainly adsorbed on the photocatalyst surface in the pure H_2O solution with the bubbling of CO_2 because of the low pH. The high concentration of carbon-related species on the surface of the photocatalyst will help suppress the formation of H_2 and promote the formation of CO during the photocatalytic conversion of CO_2 . This indicates that HCO_3^- functions as a buffer for supplying CO_2 on the surface of Ag/SrNb₂O₆ and adjusts the pH of the reaction solution to suppress the photocatalytic conversion of H^+ to H_2 and promote the photocatalytic conversion of CO_2 to

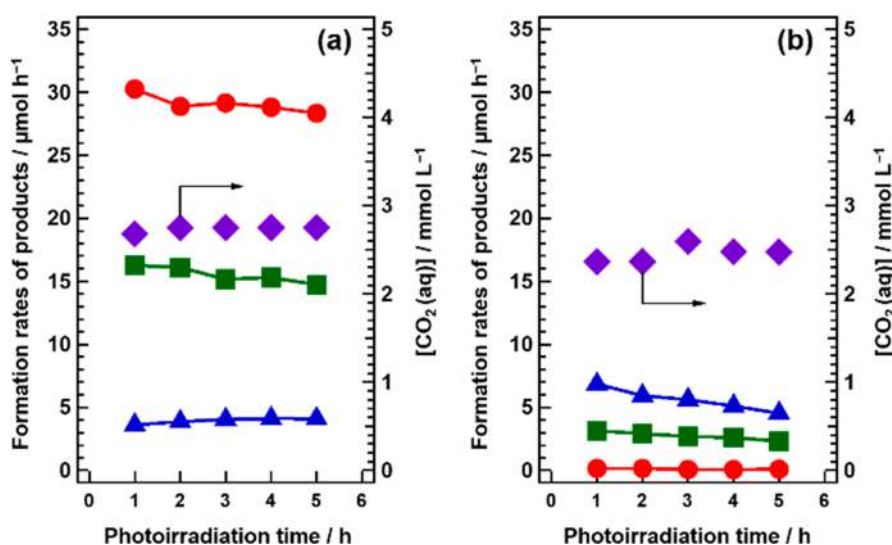
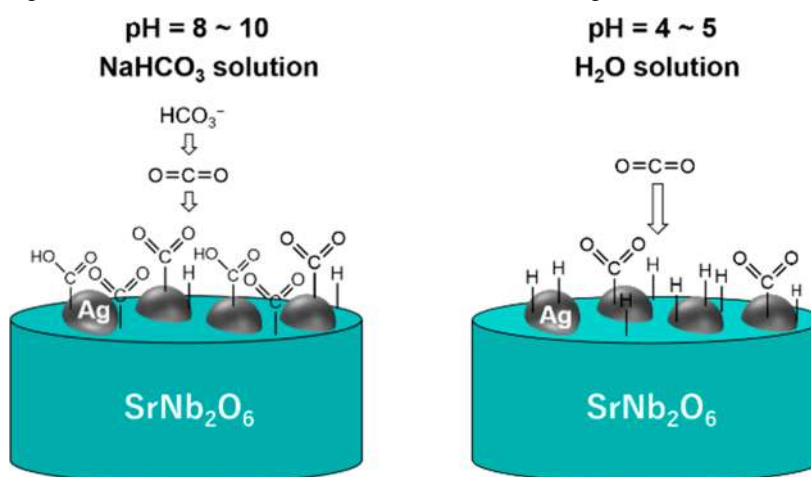


Figure 7. Formation rates of CO (red circle), H₂ (blue triangle), O₂ (green square), and [CO₂(aq)] (purple diamond) in the photocatalytic conversion of CO₂ in (a) an aqueous solution of NaHCO₃ and (b) pure H₂O solution. Photocatalyst: 1.0 wt % Ag/SrNb₂O₆ (0.5 g); reaction solution volume: H₂O (1.0 L); gas flow rate: 30 mL min⁻¹; light source: 400-W high-pressure Hg lamp.

Scheme 1. Adsorbed Species on the Surface of the Photocatalyst during the Photocatalytic Conversion of CO₂ in NaHCO₃ Solution with the Bubbling of Ar and in Pure H₂O Solution with the Bubbling of CO₂



CO. (2) From Figure 7b and Entry 1 in Table 1, we can observe that the formation rate of H₂ is quite low in the solution in pure H₂O. However, the formation rate of H₂ is significantly improved upon adding some additives such as NaOH, Na₂CO₃, or Na₂HPO₄/NaH₂PO₄ mixture (Entries 5–7 in Table 1). Some groups have reported that alkaline hydroxides or carbonates can inhibit the backward reaction in the photocatalytic splitting of water.^{34–38} Arakawa et al.^{39,40} reported that HCO₃⁻ can be activated by the photogenerated holes in the water splitting reaction and form peroxycarbonate, which easily decomposes into O₂ and CO₂ via holes under photoirradiation. They speculated that the presence of HCO₃⁻ easily desorbed the generated O₂ from the photocatalyst surface, thereby suppressing the backward reaction for water splitting (H₂ + O₂ → H₂O). Herein, we consider that the HCO₃⁻ species has a similar effect on the photocatalytic conversion of CO₂: the rapid desorption of O₂ from the photocatalyst surface is beneficial for inhibiting the backward reaction in the photocatalytic conversion of CO₂ (CO + O₂ → CO₂). The backward reaction was carried out in the same

reactor as the photocatalytic conversion of CO₂, with the simultaneous bubbling of CO and O₂ into a solution in ultrapure water. We then detected the formation rate of gaseous CO₂ under photoirradiation. As shown in Figure S10, the amounts of CO and O₂ obviously decreased, and CO₂ was formed in the H₂O solution under photoirradiation, while the bubbled CO and O₂ did not decrease and no CO₂ evolved in the NaHCO₃ solution, indicating that the backward reaction in the photocatalytic conversion of CO₂ was significantly suppressed in the presence of HCO₃⁻ species. Based on these two possibilities, we suggest that the presence of HCO₃⁻ can significantly improve the photocatalytic activity and selectivity toward CO evolution in the conversion of CO₂ in aqueous solutions containing bicarbonate ions.

CONCLUSION

We proposed a strategy of using bicarbonate as the carbon source in the photocatalytic conversion of CO₂ over Ag/SrNb₂O₆. The selectivity toward CO evolution was found to be higher than 87% for the photocatalytic conversion of CO₂ in

the presence of various bicarbonate salts added as additives. Notably, the formation rate of CO was as high as $287 \mu\text{mol h}^{-1}$ with selectivity toward CO evolution of higher than 94.1% when NH_4HCO_3 was used as an additive under the bubbling of Ar instead of CO_2 . The formation rate of CO showed a good correspondence with $[\text{CO}_2(\text{aq})]$, indicating that $\text{CO}_2(\text{aq})$ obtained by the dissociation of HCO_3^- was the actual reactant in the photocatalytic conversion of CO_2 . The following possible roles of HCO_3^- were also proposed: (1) It functions as a buffer for supplying CO_2 on the surface of Ag/SrNb₂O₆, which increases the concentrations of carbon-related species on the photocatalyst surface, and (2) it inhibits the backward reaction in the photocatalytic conversion of CO_2 by accelerating the desorption of O_2 from the Ag/SrNb₂O₆ surface. In conclusion, the presence of HCO_3^- can significantly enhance the photocatalytic activity and selectivity toward CO evolution in the conversion of CO_2 in aqueous solutions. Using bicarbonate as a carbon source for the photocatalytic conversion of CO_2 over Ag/SrNb₂O₆, the photocatalytic efficiency and utilization of CO_2 were significantly improved. We believe this study provides useful insights toward the practical application of the photocatalytic conversion of CO_2 to other feedstocks.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acsaem.9b00093.

Reactor setup, calculation details for the photocatalytic reactions in NH_4HCO_3 solution, contrast experiments for the photocatalytic conversion of CO_2 , calculated concentrations of carbon-related species, FTIR spectra of SrNb₂O₆, SEM images of Ag/SrNb₂O₆, cycle test, dependence of H_2 evolved and pH with NaHCO_3 concentration, yield of CO at different $[\text{NaHCO}_3]$, and backward reaction for the photocatalytic conversion of CO_2 in H_2O and NaHCO_3 solution (PDF)

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Notes

The authors declare no competing financial interest.

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